PROOF OF CONCEPT (POC) REPORT

Repeatable Government Services Platform

A Configuration-Driven Foundation for Whole-of-Government Digital Service Delivery

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Submitted to:
Information and Communication Technology Authority (ICTA)

Project Title:
Repeatable Government Services Platform
(Production-Centric Proof of Concept)

Purpose of the POC:
To validate the technical, operational, and governance feasibility of delivering government services through a **single, repeatable, configuration-driven platform**, capable of scaling across Ministries, Departments, and Agencies (MDAs) without recurring custom software development.

Scope of the POC:

- Mapping and onboarding of **61 MDAs**
- Configuration and execution of **132 government services** (G2C, G2B, G2G)
- Validation of service catalogue, workflow engine, RBAC, auditability, and interoperability readiness
- Assessment of production readiness, scalability, and sustainability

Date:
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Status:
Completed – Proof of Concept Phase

Report Classification:
Internal Government Use – Policy, Planning, and Decision Support

1. Executive Summary

The **Repeatable Government Services Platform Proof of Concept (POC)** demonstrates a fundamental shift in how Government of Kenya digital services can be designed, delivered, and scaled. Rather than continuing the historical pattern of building **stand-alone, service-specific systems**, the POC validates a **configuration-driven, production-centric model** where government services are defined once and executed repeatedly across Ministries, Departments, and Agencies (MDAs).

The core problem addressed by this POC is **structural**, not technical. Kenya's digital government landscape is characterized by fragmented platforms, duplicated investments, inconsistent user experiences, and high long-term operational costs. Each new digital service often results in a new system, new vendor dependency, and new integration complexity. This approach is unsustainable given the scale of government service delivery and the fiscal realities of recurring operational expenditure.

The POC establishes that **government services themselves—not systems—should be the primary unit of digital transformation**. By modeling services as standardized, machine-readable definitions, the platform decouples service design from software development. This enables MDAs to onboard, modify, and scale services through configuration rather than custom code, significantly reducing time-to-delivery and lifecycle costs.

During the POC phase, the platform successfully mapped **61 MDAs** and onboarded **132 distinct government services**, spanning citizen services (G2C), business services (G2B), and inter-governmental services (G2G). These services vary widely in complexity—from simple document requests to multi-step approval workflows—demonstrating that the repeatable services algorithm is capable of handling real-world government diversity without customization.

The POC further confirms that a **single, shared execution engine** can reliably support:

- Dynamic service workflows
- Automatic form generation
- Role-based access control for officers and supervisors
- SLA monitoring and auditability
- Secure handling of Personally Identifiable Information (PII)

Importantly, the platform was designed and tested as a **production-ready foundation**, not a conceptual prototype. Architectural decisions emphasize high availability, interoperability, security compliance, and long-term maintainability. The system is deployment-agnostic and compatible with Government Private Cloud or approved commercial cloud environments.

From a governance perspective, the POC introduces a **national Service Catalogue model** that gives ICTA visibility into what services exist, who owns them, how they are delivered, and their level of digital maturity. This provides a practical enforcement mechanism for Whole-of-Government standards, API-first mandates, and digital investment rationalization.

The results of the POC demonstrate that Kenya can transition from **project-based digitization** to a **platform-based service delivery model**—one that is scalable to thousands of services, resilient to institutional change, and ready for future capabilities such as artificial intelligence and predictive analytics.

The POC therefore provides ICTA with a validated blueprint to move from fragmented digital initiatives toward a **coherent, repeatable, and sustainable national digital services infrastructure**.

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2. Introduction & Strategic Alignment

2.1 Background and Problem Statement

The Government of Kenya delivers thousands of public services across national and county levels, cutting across citizen services (G2C), business services (G2B), and inter-governmental services (G2G). While significant progress has been made in digitising selected high-impact services, the overall digital service landscape remains **fragmented, uneven, and costly to sustain**.

Historically, digital transformation efforts have followed a **project-centric model**, where individual MDAs procure and implement standalone systems to digitise specific services. Although this approach has delivered localized gains, it has also produced several systemic challenges:

- **Duplication of digital platforms** performing similar functions (forms, workflows, payments, notifications).
- **Inconsistent service experiences** for citizens and businesses across MDAs.
- **Vendor lock-in and escalating OPEX**, particularly for hosting, licensing, and support.
- **Limited interoperability**, requiring manual verification and paper-based inter-MDA dependencies.
- **Weak central visibility** into service inventories, maturity levels, and compliance status.

As a result, the cost and complexity of digitising each additional service increases over time, while the ability to govern and standardize service delivery decreases.

2.2 Strategic Need for a Repeatable Services Model

Kenya's public sector does not suffer from a lack of digital ambition, but from the **absence of a repeatable execution model**. With an estimated **4,000–6,000 government services**, it is neither fiscally nor operationally viable to continue digitising services on a one-system-per-service basis.

The strategic requirement, therefore, is not another digital platform, but a **shared national capability** that allows:

- Services to be **defined once and reused many times**
- MDAs to focus on **policy and mandate**, not software engineering
- Digital investments to compound in value rather than fragment

The Repeatable Government Services Platform POC was initiated to test whether such a model is **technically feasible, operationally practical, and institutionally governable**.

2.3 Alignment with National Digital Priorities

The POC is directly aligned with Kenya's national digital transformation agenda and supports the strategic mandate of **Information and Communication Technology Authority** as the custodian of Whole-of-Government ICT standards.

a) National Digital Transformation Strategy

The platform advances the Strategy's core objectives by:

- Promoting **shared digital infrastructure** over siloed systems
- Enabling **end-to-end digital service delivery**
- Reducing duplication and optimizing public ICT expenditure
- Supporting inclusive access channels (web, mobile, assisted service points)

b) Whole-of-Government Architecture (WoGA)

The POC operationalizes WoGA principles by introducing:

- A **common service execution layer** across MDAs
- Standardized service definitions and workflows
- Central governance with decentralized service ownership

This ensures interoperability by design, rather than through after-the-fact integrations.

c) Digital Public Infrastructure (DPI)

The Repeatable Services Platform functions as a **service-layer DPI**, sitting above foundational registries and shared utilities such as identity, payments, and notifications.

It is designed to integrate seamlessly with:

- National identity and civil registration systems
- Revenue and licensing registries
- Messaging and payment platforms

This positions the platform as an **orchestration layer**, not a replacement for existing systems.

2.4 Strategic Outcomes for ICTA and MDAs

The POC demonstrates a path toward measurable national outcomes:

Strategic Objective	Outcome Enabled by the POC
Reduce ICT duplication	Single execution engine for all services
Improve service consistency	Standard service definitions and UX
Strengthen governance	Central Service Catalogue and auditability
Lower lifecycle costs	Configuration over custom development
Enable future innovation	AI-ready, API-first architecture

For MDAs, the model lowers the barrier to digitisation while preserving **institutional autonomy and legal mandates**.

2.5 Why a POC Was Necessary

Given the scale and sensitivity of government service delivery, a theoretical or conceptual approach was insufficient. The POC was deliberately designed to:

- Use **real MDA data**, not hypothetical services
- Model **actual workflows and approval chains**
- Surface real-world constraints (PII, approvals, dependencies)
- Validate performance under realistic conditions

The findings presented in this report are therefore **evidence-based**, not aspirational.

2.6 Transition from POC to National Capability

This POC is not an end state. It is a **decision-support instrument** for ICTA to determine:

- Whether a repeatable services model can replace fragmented digitisation
- How national service governance can be practically enforced
- Which services should be prioritized for pilot and rollout

The results confirm that the model is viable and ready to progress into a **controlled pilot phase** with selected high-impact services.

2.2 Strategic Objective
The primary goal of this POC was to build a **production-ready foundation** for a repeatable services framework. This aligns with the National Digital Transformation Strategy by providing a single, reusable engine for all government G2C, G2B, and G2G interactions.

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3. The "Repeatable Services" Solution

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3.1 Core Concept

The central innovation of this project is the **Repeatable Services Algorithm**—a service execution model that treats **government services as configurable products**, not custom-built software systems.

In conventional digital government implementations, each new service is delivered by:

- Designing a bespoke workflow
- Building custom forms
- Writing service-specific business logic
- Integrating individually with external systems

This results in **high cost, slow delivery, and limited scalability**.

The Repeatable Services Platform replaces this model with a **single execution engine** that interprets **machine-readable service definitions** expressed in JSON or YAML. These definitions describe *what* a service is, *how* it should flow, *who* is involved, and *what data is required*—without embedding this logic in code.

As a result:

> Adding a new service becomes a **configuration exercise**, not a software development project.

3.2 What Makes a Service "Repeatable"

A service is considered repeatable when it satisfies three conditions:

1. **Standardized Structure**
   Every service—regardless of sector or MDA—follows the same core definition model (inputs, workflow, roles, outputs, SLAs).

2. **Engine-Driven Execution**
   The platform executes services dynamically based on configuration, rather than hard-coded logic.

3. **Decoupling from Institutions**
   The same execution logic applies whether the service belongs to a Ministry, State Department, Agency, or County Government.

This abstraction is what allows the platform to scale from **dozens to thousands of services** without exponential cost growth.

3.3 Key Functional Capabilities

a) Dynamic Workflow Engine

The workflow engine executes **multi-step service lifecycles** using declarative definitions rather than fixed process flows.

Typical stages include:

- Request submission
- Automated validation
- Officer review
- Supervisor approval
- Fulfilment and notification

Workflows can be:

- **Simple** (single approval)
- **Hierarchical** (officer → supervisor → director)
- **Cross-functional** (multiple departments)

All variations are handled by the same engine, driven entirely by configuration.

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b) Universal Form Generator

Citizen- and officer-facing forms are generated automatically from the service definition.

This capability ensures:

- Consistent user experience across MDAs
- Built-in validation rules
- Faster service onboarding
- Reduced form design errors

Changes to requirements (e.g. adding a document or field) are applied through configuration and immediately reflected in the user interface.

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c) Role-Based Access Control (RBAC)

The platform enforces **institutional roles**, not system-specific permissions.

Standard roles include:

- Citizen / Applicant
- Processing Officer
- Supervisor / Approver
- System Administrator

Each role is mapped to:

- Allowed actions
- Visibility of data
- Escalation authority

This ensures accountability, auditability, and compliance with public service operational structures.

d) Authoritative Registry Integration

The solution is designed for **real-time validation against authoritative government registries**, eliminating manual verification and paper-based dependencies.

The POC demonstrates readiness to integrate with:

- Integrated Population Registration System (identity verification)
- Kenya Revenue Authority (tax compliance)
- Business Registration Service (business data)

This architecture supports an **API-first government mandate**, enabling services to verify data at source rather than relying on uploaded documents.

3.4 Why This Is Not “Just Another Platform”

The Repeatable Services Platform does **not** compete with existing systems such as registries, ERPs, or sector platforms. Instead, it acts as a **service orchestration layer** that:

- Consumes authoritative data
- Applies policy and workflow rules
- Delivers services consistently across channels

It is intentionally:

- **Technology-agnostic**
- **Vendor-neutral**
- **Policy-driven**

This makes it suitable as a **national shared capability**, rather than an MDA-owned application.

3.5 Proof from the POC

During the POC:

- **132 services** were onboarded
- **61 MDAs** were mapped
- No service required custom workflow code
- The same engine executed all service types

This empirically demonstrates that the repeatable services model is **not theoretical**, but operationally viable under real government conditions.

3.6 Strategic Significance

The Repeatable Services Solution establishes a foundation where:

- ICTA governs *standards and execution*
- MDAs govern *policy and mandate*
- Citizens experience *consistent, predictable services*

It transforms digital government from a collection of projects into a **scalable national capability**.

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4. Technical Architecture

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4.1 Architectural Design Principles

The Repeatable Government Services Platform POC was designed using a **production-centric, technology-agnostic architecture**. The objective was not to showcase a specific technology stack, but to validate an architectural pattern that can be **safely adopted as a national shared platform**.

The architecture is guided by the following principles:

- **Separation of concerns:** Service logic, presentation, data, and infrastructure are fully decoupled.
- **Configuration over code:** Core service behavior is driven by declarative definitions, not hard-coded workflows.
- **API-first by default:** All functionality is exposed through secure, documented APIs.
- **Scalability and resilience:** Designed to support high transaction volumes without degradation.
- **Security and compliance by design:** Aligned with public sector data protection and audit requirements.

4.2 High-Level Architecture Overview

At a high level, the platform is composed of five logical layers:

1. Service Execution Layer
2. Presentation Layer
3. Data Persistence Layer
4. Asynchronous Processing Layer
5. Deployment & Infrastructure Layer

Each layer can evolve independently without disrupting service delivery.

4.3 Backend Layer – Service Execution Engine

The backend consists of an **API-first Application Server** that functions as the **core service execution engine**.

Its responsibilities include:

- Interpreting service definitions (JSON/YAML)
- Executing workflow steps and state transitions
- Enforcing business rules and validations
- Managing role-based permissions
- Coordinating interactions with external systems

All service behavior—forms, workflows, roles, and SLAs—is resolved dynamically at runtime. This ensures that introducing or modifying a service does **not require redeployment of application code**.

The backend exposes standardized RESTful endpoints, enabling:

- Multiple frontend clients
- Secure inter-MDA integrations
- Future AI agent consumption

4.4 Frontend Layer – Dynamic Service Interfaces

The frontend layer is implemented as a **reactive client application** that renders user interfaces dynamically based on service configuration metadata.

Key characteristics include:

- No service-specific UI code
- Real-time validation and feedback
- Support for multiple user roles (citizen, officer, supervisor)

Because the frontend consumes the same APIs as other channels, the architecture supports:

- Web portals
- Mobile applications
- Assisted service kiosks
- USSD and SMS gateways (through adapters)

This approach ensures **channel consistency** while allowing MDAs to expand access without re-engineering services.

4.5 Data Persistence – Transactional Integrity and Auditability

The platform uses an **enterprise-grade relational database** as its primary data store.

This design choice was driven by:

- The need for **strong transactional guarantees (ACID)**
- High-volume, structured service data
- Long-term audit and compliance requirements

The data model supports:

- Complete service request histories
- Immutable audit logs

- SLA and performance metrics
- Secure segregation of PII-sensitive data

This ensures compliance with the **Data Protection Act, 2019** and public sector record-keeping standards.

4.6 Asynchronous Processing and Background Orchestration

Government services frequently involve **long-running and non-interactive tasks** such as:

- External registry validations
- Notifications (SMS, email)
- SLA tracking and escalations
- Batch reporting

To prevent performance bottlenecks, these activities are handled by an **asynchronous task management system**.

This design:

- Improves responsiveness for users
- Enables horizontal scaling of background workers
- Supports future automation and AI-driven processes

The asynchronous layer is critical for handling national-scale transaction volumes without impacting front-office service delivery.

4.7 Deployment and Infrastructure Strategy

The platform is deployed using a **containerized orchestration approach**, based on Docker-compatible standards.

This ensures that the solution is:

- **Environment-agnostic**

- Easily portable across infrastructure providers
- Consistent across development, staging, and production

The architecture supports deployment on:

- Government Private Cloud
- Approved commercial cloud platforms
- Hybrid cloud environments

Infrastructure scaling (compute, storage, networking) can be performed independently of application logic, allowing ICTA to optimize cost, resilience, and performance.

4.8 Security and Compliance Considerations

Security is enforced across all architectural layers through:

- API authentication and authorization
- Role-based access control
- Secure data storage and transmission
- Comprehensive logging and audit trails

The architecture supports alignment with:

- National cybersecurity guidelines
- Data protection regulations
- ICTA security standards and assessments

4.9 Architectural Readiness for Scale and Evolution

The POC architecture was validated as:

- **Production-ready**

- **Scalable to thousands of services**
- **Extensible for AI and advanced analytics**

It is deliberately designed to accommodate:

- New service categories
- Additional MDAs and counties
- Future shared services (payments, identity, messaging)

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5. Data Insights & MDA Landscape Analysis

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5.1 Data Sources and Methodology

The analysis presented in this section is based on **primary data extracted from ICTA Digital Readiness Assessments**, supplemented by service inventories collected during the POC configuration exercise.

The objective of this analysis was to:

- Validate whether the repeatable services model can operate across **heterogeneous MDAs**
- Identify structural constraints affecting national service digitisation
- Ground architectural and policy recommendations in **empirical evidence**

The dataset covered **49+ MDAs**, including Ministries, State Departments, and Agencies with varying mandates, service volumes, and digital maturity levels.

5.2 Digital Maturity Distribution Across MDAs

A key finding of the analysis is the **uneven distribution of digital maturity** across government.

High-volume agencies such as:

- Kenya Revenue Authority
- Kenya National Examinations Council
- Business Registration Service

operate at **Level 5 maturity**, characterized by:

- Fully digitised service workflows
- API-accessible authoritative registries
- High transaction throughput
- Established cybersecurity practices

In contrast, a significant number of State Departments and regulatory bodies remain at **Levels 1–2**, where:

- Over **90% of records are paper-based**
- Service requests rely on physical forms
- Digital systems, if present, are siloed and non-interoperable

This disparity creates **systemic inequity in service delivery** and amplifies integration challenges across MDAs.

5.3 Service Density and Complexity

The POC identified **132 distinct government services** across the participating MDAs. These services span a wide spectrum of complexity:

Service Type	Characteristics
Low Complexity	Single-step requests, document issuance
Medium Complexity	Multi-step reviews, officer approvals
High Complexity	Cross-MDA dependencies, external validations

Notably, service complexity does not correlate with MDA size or budget. Smaller MDAs often manage **highly complex regulatory services** without adequate digital tooling.

This reinforces the need for a **shared execution platform** that abstracts complexity away from individual institutions.

5.4 PII Exposure and Data Protection Risk

A critical insight from the analysis is the **pervasive handling of Personally Identifiable Information (PII)** across almost all services assessed.

Examples include:

- Identity data
- Contact information
- Financial records
- Biometric-linked identifiers

Despite this, many MDAs reported:

- Data Protection policies in **draft form**
- Limited enforcement mechanisms
- Inconsistent audit logging
- Manual handling of sensitive documents

This represents a **systemic compliance risk**, particularly under the **Data Protection Act, 2019**.

The POC architecture directly addresses this risk by enforcing:

- Centralized access control
- Immutable audit trails
- Standardized data classification at service definition level

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5.5 Structural Bottlenecks in Service Delivery

The analysis identified **five recurring bottlenecks** that affect service delivery across MDAs, regardless of mandate or size.

1. Manual Forms and Paper-Based Processes

Paper remains the dominant medium for service initiation, approvals, and record-keeping. This results in:

- Long turnaround times
- Lost or duplicated files
- Limited traceability

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2. Inter-MDA Dependencies Without Integration

Many services depend on verification from other MDAs, yet lack real-time system-to-system integration. This forces:

- Manual letters and memos
- Physical file movement
- Delayed decision-making

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3. Approval Chain Delays

Multi-level approval processes are often executed manually, leading to:

- Bottlenecks at supervisory levels
- Lack of SLA enforcement
- No visibility into service backlogs

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4. Budget Constraints and OPEX Challenges

MDAs struggle to sustain:

- Recurring cloud costs
- Software licensing
- Security monitoring

This results in stalled or abandoned digital systems after initial implementation.

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5. Digital Skills Gap

There is a shortage of:

- Cloud infrastructure expertise
- Cybersecurity skills
- API and integration specialists

This further reinforces the case for **centralized platforms managed at national level**.

5.6 Implications for National Digital Strategy

The insights from this analysis confirm that:

- Fragmented digitisation will continue to increase risk and cost
- MDA-by-MDA solutions cannot scale sustainably
- Central governance without shared infrastructure is ineffective

The Repeatable Government Services Platform directly addresses these constraints by:

- Standardizing service execution
- Reducing technical burden on MDAs
- Centralizing compliance and security controls
- Enabling interoperability by default

5.7 Evidence of Platform Fit

The fact that the platform successfully onboarded **132 services across diverse MDAs**, despite these constraints, provides strong evidence that the repeatable services model is:

- Institution-agnostic
- Resilient to maturity disparities
- Suitable for national rollo

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6. Functional Results & Deliverables

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The Proof of Concept delivered a **fully integrated, end-to-end functional stack** that validates the Repeatable Government Services Platform under realistic government service conditions. Each component was implemented to demonstrate not only usability, but **institutional fit, auditability, and production readiness**.

6.1 Citizen Portal – End-to-End Service Access

The POC includes a citizen-facing portal that simulates the **full service lifecycle**, from application to fulfilment.

Key capabilities:

- Dynamic service discovery from the Service Catalogue
- Auto-generated application forms
- Document submission and validation
- Real-time service status tracking
- Notifications at key workflow stages

The portal demonstrates that citizens can experience **consistent service journeys** regardless of the owning MDA, reinforcing the Whole-of-Government service model.

6.2 Officer Workbench – Service Processing and Control

The Officer Workbench provides a **role-aware operational interface** for public officers responsible for service processing.

Capabilities include:

- Queued service requests by role and responsibility
- Validation and review of submissions
- Structured decision-making (approve, reject, escalate)
- Inline access to service definitions and requirements
- Full traceability of actions taken

This component proves that **operational complexity is handled by the platform**, not individual MDAs or officers.

6.3 Supervisor Monitoring – Oversight and Performance Management

Supervisors and managers are provided with **real-time visibility** into service operations.

Delivered functionality includes:

- Live service volume metrics
- SLA compliance tracking
- Bottleneck identification
- Escalation oversight
- Historical performance trends

This enables **evidence-based management**, replacing anecdotal or manual reporting methods.

6.4 Audit Module – Compliance and Accountability

The Audit Module implements **immutable, system-enforced logging** of every service interaction.

Captured events include:

- Service request creation
- Data access and modification
- Officer and supervisor decisions
- Workflow transitions
- System-triggered actions

This ensures:

- End-to-end accountability
- Non-repudiation of actions
- Alignment with national data protection and records management requirements

The audit capability is **inherent to the platform**, not an add-on feature.

6.5 Consolidated Service Matrix – National Visibility

One of the most significant POC outputs is the **Consolidated Service Matrix**, generated directly from platform configurations.

This deliverable provides:

- A mapping of **132 services** to their respective MDAs
- Classification by service type (G2C, G2B, G2G)
- Digital maturity levels
- Workflow complexity indicators
- Data sensitivity flags

The matrix serves as:

- A **national service inventory**

- A baseline for digitisation prioritisation
- A governance tool for ICTA oversight
- An evidence base for budgeting and planning

6.6 Validation of Repeatability

Collectively, these deliverables confirm that:

- Services can be onboarded without custom development
- Multiple user roles can operate on the same platform
- Governance, performance, and compliance are enforced centrally
- The platform functions under real-world institutional constraints

This moves the solution beyond conceptual validation into **operational proof**.

6.7 Readiness for Pilot Deployment

The functional stack delivered in the POC is sufficient to support:

- Controlled pilots with selected high-impact services
- Multi-MDA onboarding
- Performance benchmarking
- Security and compliance assessments

No architectural refactoring is required to proceed to pilot phase.

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7. Recommendations for Production Scale-up

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The POC confirms that the Repeatable Government Services Platform is technically viable and operationally effective. To transition from POC to a **national production capability**, targeted policy, funding, and governance actions are required.

The following recommendations are prioritized to ensure sustainability, interoperability, and long-term value for public investment.

7.1 Sustainability: Transition from CAPEX to OPEX-Driven Service Models

Current government digitisation funding is predominantly **project-based**, focusing on one-time system development costs (CAPEX). While this approach enables initial deployment, it consistently fails to support:

- Ongoing cloud hosting
- Cybersecurity monitoring
- System updates and compliance maintenance
- Performance optimization and scaling

Recommendation

ICTA should formally adopt a **service-based funding model**, where national digital platforms are funded through recurring **Service Level Agreements (SLAs)** covering:

- Cloud infrastructure hosting
- Security operations (SOC monitoring, vulnerability management)
- Platform maintenance and upgrades
- Disaster recovery and business continuity

Strategic Impact

- Predictable and controllable OPEX
- Reduced risk of system abandonment after launch

- Improved service reliability and uptime
- Alignment with modern cloud and shared-services economics

7.2 Enforce an API-First Mandate for Authoritative Registries

A major cause of service delays identified in the POC is the reliance on **manual inter-MDA verification**, often implemented through letters, memos, or physical file movement.

Recommendation

ICTA should issue and enforce an **API-First Government Directive** requiring MDAs that manage authoritative datasets to:

- Expose secure, documented RESTful APIs
- Support real-time or near-real-time data validation
- Publish service contracts and access rules
- Integrate with national service orchestration platforms

Target datasets include:

- Identity and civil registration
- Business registration
- Land and property records
- Licensing and regulatory registries

Strategic Impact

- Elimination of manual inter-MDA dependencies
- Faster service turnaround times
- Reduced fraud and data inconsistency
- Foundation for Whole-of-Government interoperability

7.3 Phased Integration of Artificial Intelligence Capabilities

The POC architecture was deliberately designed to be **AI-ready**, ensuring that future intelligence capabilities can be integrated without re-engineering the platform.

Recommendation

ICTA should adopt a **phased AI integration roadmap**, focusing on high-impact, low-risk use cases:

Phase 1: Assisted Automation

- Automated first-level document validation
- Duplicate detection and completeness checks
- Rule-based eligibility pre-screening

Phase 2: Citizen Interaction

- Natural language virtual assistants for service guidance
- Multilingual support for inclusive access
- Assisted service navigation for complex workflows

Phase 3: Predictive Intelligence

- SLA breach prediction
- Service demand forecasting
- Workflow bottleneck detection and optimization

Strategic Impact

- Reduced administrative burden on officers
- Improved citizen experience
- Data-driven service management
- Gradual, governed adoption of AI in public service

7.4 Governance and Ownership Model

To ensure institutional clarity and sustainability:

- ICTA should retain **platform ownership and governance**
- MDAs should retain **service ownership and policy authority**
- Service onboarding should follow a standardized approval workflow
- Security, compliance, and interoperability standards should be centrally enforced

This preserves decentralised mandates while ensuring **national consistency**.

7.5 Recommended Pilot Approach

Rather than attempting immediate national rollout, the following pilot strategy is recommended:

- Select **3–5 high-volume, high-impact services**
- Prioritize services with cross-MDA dependencies
- Measure performance, cost, and user experience
- Use pilot results to refine rollout and funding models

8. Conclusion

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The **Repeatable Government Services Platform Proof of Concept (POC)** has successfully demonstrated that a **configuration-driven, service-centric model** is both **technically viable** and **strategically superior** to the traditional approach of siloed system development.

Through the onboarding of **132 real government services across 61 MDAs**, the POC confirms that government services—regardless of complexity, ownership, or maturity level—can be delivered through a **single, shared execution platform** without recurring software development. This directly addresses long-standing challenges of duplication, interoperability, escalating costs, and inconsistent service delivery across government.

Beyond technical validation, the POC establishes a **new operating model for digital government**:

- ICTA governs standards, execution, and shared infrastructure
- MDAs retain policy authority and service ownership
- Services are digitised as configurable products rather than bespoke systems
- Compliance, auditability, and performance management are built in by design

The architecture and functional stack delivered under this POC are **production-ready** and aligned with national priorities, including Whole-of-Government Architecture, Digital Public Infrastructure, and sustainable ICT financing models. The platform is also future-proofed, with native support for API-driven interoperability and phased adoption of artificial intelligence capabilities.

Critically, this POC shifts the national conversation from *"Which system should we build next?"* to *"Which services should we onboard next?"*—a change that is essential for scaling digital government across thousands of services.

The evidence presented in this report supports a clear next step:
transitioning the platform into a controlled pilot phase, focused on high-impact services such as Civil Registration, Housing, or other cross-MDA workflows. This will allow ICTA to validate operational performance at scale, refine governance mechanisms, and establish the foundation for national rollout.

In conclusion, the Repeatable Government Services Platform provides ICTA with a **validated blueprint for sustainable, whole-of-government digital service delivery**—one that reduces cost, increases resilience, and places citizens and service outcomes at the center of digital transformation in Kenya.

Appendices Attached

- **Government_Services_List.csv** – Full mapping of services to MDAs and maturity levels
- **System Design Documents** – Architecture diagrams and ERD
- **Algorithm Documentation** – Pseudo-code and execution logic